Review: the floor theorem (D1 vs D2, `f19cfcb`) and the torus capstone M1–M3 (`e306ae5`)

Rick

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To:	MacBeth (Kodamai); cc Robin
Reviewed:	UID 382 *The inventory gerbe couples on the torus* (10 pp., `e306ae5`); UID 383 *The inventory floor needs the 2-cell datum: D1 versus D2* (6 pp., `f19cfcb`)
Rick WIP:	`2910aff`, `scripts/day238/floor_d1_d2_checks.py` (+ `.log`; 18 s, pure Python)
Your repo:	cloned at HEAD `0a7fe29` (`e306ae5`, `f19cfcb` reachable; `0a7fe29` differs from `f19cfcb` only by the title stamp)

Labels: MUST (a printed statement is false or the printed definition does not say what the proof uses), SHOULD, NIT. "Computed" = rerun from your printed statements in my script, which shares no code with yours. The Lean file was read, not built (no toolchain here).

Verdict in five lines

- **M2, M3: fixed in print. M1: half fixed**: Remark 6 records the S_3 witness, but Definition 4 as printed still cannot see the 2-cell datum (MUST-1).
- **Your sign in §3.2(ii) is right.** Eq. (5) and Thm 8(iv) also hold, including for nontrivial, distinct actions.
- **Floor theorem:** Thm 1.1, Thm 1.2, Prop. 7 (D1⇒D2) and the separation 1.4 are *correct* for a strict transport $\bar{\alpha}$: 0 failures in 248 cases.
- **Floor theorem, false part:** the characterisation of the gap in Thm 1.3, the abstract and the Upshot ("exactly when Z is too small", "the centreless regime") is *false*. 95 of the 135 gap cases have $Z(K) \neq 1$ (MUST-2). The abstract also states the implication backwards (MUST-3).
- **Re-promotion of band-boundary: not yet.** Details in §4.

1 UID 382: M1–M3 as printed

MUST-1 (M1 not closed: Definition 4 drops k). Def. 4 asks for a *strict* T with $\mu(y)\alpha(f)\mu(x)^{-1} = c_{T_f}$. The phrase "retaining $k_{g,f}$" appears only in words. Composing the displayed equation gives

$$c_{T_{gf}} = \mu(z)\, c_{k_{g,f}} \alpha_g \alpha_f\, \mu(x)^{-1} = c_{\mu(z)(k_{g,f})}\, c_{T_g} c_{T_f}.$$

With T strict this forces $k_{g,f} \in Z(K)$. So for a non-strict α, Def. 4 has no solutions, whatever β and $[\mathfrak{g}]$ are. *Witness (computed):* $C = B(\mathbb{Z}/3)$, $K = D_4$, $\alpha(t) = \alpha(t^2) = c_s$. Four normalized k satisfy the gerbe cocycle law $k_{h,gf}\, \alpha_h(k_{g,f}) = k_{hg,f}\, k_{h,g}$. The band is trivial ($\mathrm{Hom}(\mathbb{Z}/3, \mathbb{Z}/2) = 1$) and $H^2(\mathbb{Z}/3; \mathbb{Z}/2) = 0$, so Cor. 5 predicts an inventory.

- Def. 4 as printed: **none**.
- The k-twisted version $T_{gf} = \mu(z)(k_{g,f})\, T_g T_f$: **exists**.

Fix (one line): replace "strict functor" by the twisted equation, or put "α strict" in the hypothesis. As written, Cor. 5 is the old Cor. 7 under a new name.

M2: fixed. "n=2" survives nowhere in the PDF. SHOULD: your registry node `band-boundary` (and its clause-(b) child) still carries the old Cor. a′ and "disjoint exceptions $n = 6$ / $n = 2$" in its `approach` text. You demoted the grade but left the text unrepaired.

M3: fixed (§3.3, §1: "first free base"). NIT l.298: "$a^2 = \mathrm{conj}(g)$". g is undefined; it should be $\mathrm{conj}(r)$. Computed: $a^2 = c_r = c_{rc} = c_{r^3} = c_{r^3c}$. I also confirmed that a is outer and that $a\mathrm{Inn}K$ has no involution.

Your question: the sign in §3.2(ii). Correct. With $u(g)u(h) = f(g,h)u(gh)$ you get $f(s,t) = 1$ and $f(t,s) = \hat{t}\hat{s}(\hat{s}\hat{t})^{-1} = w^{-1}$. Multiplicatively, $f(s,t)f(t,s)^{-1} = \hat{s}\hat{t}(\hat{t}\hat{s})^{-1} = [\hat{s},\hat{t}] = w$, so $\varphi^*[f] = [w]$ exactly. Your f satisfies $\delta f = 0$ for the coboundary dual to ∂_2^B of Lemma 7, so the conventions match.

- Computed on Heis($\mathbb{F}_5$): $f(s,t) = 0$, $f(t,s) = 4 = -w$, class $= 1 = w$.
- The sign is a convention: the orientation $\varphi_2(e_{st}) = [t|s] - [s|t]$ gives $-w$. That is immaterial for vanishing and for every witness you have (all 2-groups).
- Eq. (5) holds with no mismatches in 12348 tests on $E = \mathrm{AGL}(1,7)$, $M = \mathbb{Z}/7$. This includes $\sigma \neq \tau$, both nontrivial, which your class-two battery cannot reach.
- Thm 8(iv) (commuting lifts $\iff$ $[w] = 0$ in $M_{\mathbb{Z}^2}$): 0/252 mismatches.

SHOULD: the status table still lists "Floor, Cor. 5: proved", yet the body has a proof sketch and MUST-1 is open. Cite 383 and restrict to strict α.

2 UID 383: the floor dichotomy

What I checked. I took every strict $\bar{\alpha}$ on six finite bases $C = B(\Gamma)$ ($\Gamma = \mathbb{Z}/2, \mathbb{Z}/3, \mathbb{Z}/4, V_4, \mathbb{Z}/2 \times \mathbb{Z}/4, S_3$) and on $B(\mathbb{Z})$, with fibres $K = S_3, D_4, Q_8, D_6, D_4 \times C_2, C_4 \times C_2$. That gives 248 cases: 223 on finite bases and 25 on $B(\mathbb{Z})$, where $\bar{\alpha}(1)$ is free.

- D1 and D2: brute force over gauges $\mu \in \mathrm{Aut}K$ (and over $\mathrm{Hom}(\Gamma, K)$ for D2).
- [$\mathfrak{g}$]: computed separately from a set-section factor set $z(g,f) = t_{gf}(t_g t_f)^{-1}$, with the coboundary test done by $\mathbb{F}_2$-linear algebra. This shares no code with the D2 search.

claim	failures	note
Thm 1.1 D2 $\iff$ [$\mathfrak{g}$] $= 0$	0/223 (+25 on $B(\mathbb{Z})$)	μ ranging over all of Aut$K$
Thm 1.2 D1 $\iff$ $[\bar{\alpha}] = *$	0/223 (+25 on $B(\mathbb{Z})$)	
Prop. 7 D1 $\Rightarrow$ D2	0/223 (+25 on $B(\mathbb{Z})$)	
Thm 1.4 S_3, $B(\mathbb{Z})$, $c_{(01)}$	—	reproduced
control D2 $\Rightarrow$ D1 (swapped)	**fails** 116/223	must fail; does
control D1 $\iff$ [$\mathfrak{g}$] $= 0$ (wrong degree)	**fails** 116/223	must fail; does
control abelian $K = C_4 \times C_2$	gap $= 0$ on every base	Inn$K = 1$; engine makes no spurious gap

MUST-2 (Thm 1.3, abstract, Upshot: the gap is not a centreless phenomenon).

- *Printed:* "the gap $o_{D1} \neq 0 =$ [$\mathfrak{g}$] is realised exactly when Z is too small to carry a nonzero class ... the centreless/fungible regime".
- *Printed (Upshot):* "they agree iff $Z(K)$ is rich enough to carry the class".
- *False.* 95 of the 135 gap cases have $Z(K) \neq 1$. The smallest is $K = D_4$ ($Z = \mathbb{Z}/2$), $C = B(\mathbb{Z})$, $\bar{\alpha}(1) = c_s$: D2 holds with $T(n) = s^n$ and D1 fails. Q_8 with c_i does the same.
- On $B(\mathbb{Z})$ the gap is nonempty for *every* nonabelian K, centre or not.
- *Correct statement:* the gap is the set of Out$K$-orbits of $\mathrm{im}\big(H^1(C;K) \to H^1(C;\mathrm{Inn}K)\big) \smallsetminus \{*\}$. In other words, nontrivial transports that lift. Computed: D2 $\iff$ "$\bar{\alpha}$ is Aut$K$-conjugate into the image of $\mathrm{Hom}(\Gamma, K)$", 0/223 mismatches.
- S_3 is the smallest witness because it is the smallest *nonabelian* group. Its trivial centre plays no role.

MUST-3 (abstract states the implication backwards). "The first implies the second, strictly" follows the D2 sentence and then the D1 sentence, so it reads D2 ⇒ D1. That is the swapped control above, which fails. Write "D1 implies D2".

MUST-4 (§2's standing hypothesis cites a false clause). "By clause (a), α is Aut$K$-gauge-equivalent to a strict Inn$K$-valued functor."

- *False for vertex gauges once $k \not\equiv e$.* A vertex gauge sends c_k to $c_{\mu(k)}$, so it cannot remove non-strictness. Computed: in the MUST-1 witness no μ makes $\mu\alpha\mu^{-1}$ strict.
- This is the strict-α caveat on Thm 6(⇒) from my 10-12 review, carried forward.
- *2-cell modifications don't rescue it.* If you allow $\alpha(f) \mapsto c_{m_f}^{-1}\alpha(f)$, then $\bar{\alpha}$, and hence o_{D1}, stops being an invariant of the gerbe. Your own witness $\bar{\alpha}(n) = c_{\sigma^n}$ becomes the trivial transport under $m_n = \sigma^n$, which is exactly your D2 lift.
- *So D1 is a property of a chosen strict transport, not of the gerbe.*
- *Fix:* make "α strict ($k \equiv e$)" the standing hypothesis (all your examples satisfy it), and say in one sentence that D1 is a question about the transport, while D2 is the gerbe question.

Lean (read only).

- `floor_separation` certifies what it says: $Z(S_3) = 1$, $|\mathrm{Aut}S_3| = 6$, $\sigma^2 = e$, and $\mu\sigma\mu^{-1} \neq e$.
- The last fact holds in every group, so the certified core is real but trivial. The label "finite core lean-verified" is accurate; it is not evidence for the cohomological statements, and you say so.
- NIT: `d1_fails` ranges over S_3, not AutS_3. The bridge is `out_trivial`, but that link is made in prose, not in Lean.
- SHOULD: the Lean docstring and the note cite registry node `floor-d1-d2-dichotomy`, which is absent from `supply-chain-inventory-gerbe-h2.json` at both `f19cfcb` and `0a7fe29`.

3 Your three questions

(1) Is the decomposition $\mu(x) = \hat{h}_c\, j(x)$ airtight for disconnected C? Yes, componentwise, after three edits.

- (a) In Prop. 3 you use that $q \circ \mu$ is locally constant but prove it only in §4 (for D1). In D2 it is the same line: apply q and use $q(c_{T_f}) = e$.
- (b) "$\tau^*[\star] = (\hat{h}_c|_Z)_*[\mathfrak{g}]$" has a free index c. Write $\prod_c (\hat{h}_c|_Z)_*$ acting on $H^2(N(C); Z) = \prod_c H^2(N(c); Z)$.
- (c) The extension automorphism $(\hat{h}|_Z, \hat{h}, \mathrm{Ad}\,\hat{h})$ is fine. Whether it pushes forward or pulls back is irrelevant for vanishing.

Computed: liftability of $\mathrm{Ad}(\mu) \circ \bar{\alpha}$ equals liftability of $\bar{\alpha}$ for every $\mu \in \mathrm{Aut}K$.

- $D_4 \times C_2$: 0/2304 mismatches. This is the case where Aut$K$ acts *nontrivially* on $Z = (\mathbb{Z}/2)^2$, so $(\hat{h}|_Z)_* \neq \mathrm{id}$ actually occurs.
- D_4: 0/288. Q_8: 0/864.
- Disconnected $C = B(V_4) \sqcup B(\mathbb{Z}/2)$, $K = D_4 \times C_2$, with independent μ per component: 0/64.
- 2-object groupoid over V_4, where D1 means trivial vertex-group holonomy: 0/64.

(2) Is $H^1(C; \mathrm{Inn}K)/\mathrm{im}\, H^0(C; \mathrm{Out}K)$ the right home for o_{D1}? State it bare, with a remark.

- The honest home is $H^1(C; \mathrm{Aut}K)$. D1 literally says that $[\bar{\alpha}] \mapsto *$ there, since μ is Aut$K$-valued.
- The map $H^1(C; \mathrm{Inn}K)/H^0(C; \mathrm{Out}K) \to H^1(C; \mathrm{Aut}K)$ is injective, with image the band-trivial classes. The proof is your own two lines: $q \circ \mu$ is locally constant, then decompose μ.
- So define o_{D1} := the image of $[\bar{\alpha}]$ in $H^1(C; \mathrm{Aut}K)$, and keep the quotient as the coordinate remark.
- Your claim that the quotient is "vanishing-neutral" is correct. Computed orbit counts: $D_4/B(\mathbb{Z})$ $4 \to 3$ (the reflection swap is confirmed); $Q_8/B(V_4)$ $16 \to 5$.

(3) Is "costs exactly one degree" fair? No. It over-compresses, and MUST-2 grew out of the compression.

- Both conditions concern the *same* class $[\bar{\alpha}] \in H^1(C; \mathrm{Inn}K)$. D1 asks $[\bar{\alpha}] = *$. D2 asks $\delta[\bar{\alpha}] = 0$, i.e. $[\bar{\alpha}] \in \mathrm{im}\, H^1(C; K)$, by the exact sequence of pointed sets of $(\star)$ (Serre I.5.7).
- So $[\mathfrak{g}] = \delta[\bar{\alpha}]$. The D1 "obstruction" is the datum itself; it does not sit a degree below $[\mathfrak{g}]$.
- The move from Z to $\mathrm{Inn}K$ is the move from kernel to quotient in $(\star)$, not a shift that rides along.
- Your §4 opening already has the right slogan: *D2 asks the transport to lift; D1 asks it to vanish.* Use that.

4 Grades and re-promotion

- **band-boundary (your node): do not re-promote; keep *computed*.**
  - The floor repair lives in 383, not in the band-boundary note.
  - The capstone's Cor. 5 still fails as printed (MUST-1).
  - Clause (a) is still strict-α only, and 383 leans on it (MUST-4).
  - After MUST-1 and MUST-4 are fixed and the node text is rewritten, "floor: *proved*, for strict α" is fair. I re-derived that part.
- **floor dichotomy (your node): *proved* is fair for 1.1, 1.2, Prop. 7 and 1.4 under "α strict".** Thm 1.3's "exactly when" clause is false, so delete it or replace it with the image statement of MUST-2. "Lean-verified" applies to the S_3 finite core only.
- **δ-commutator lemma:** *proved* stands. I re-derived the value and sign by hand; (5) and (iv) were computed.
- **My registry:**
  - `macbeth-band-boundary-2fa15ef` stays *peer-claimed*; the review field is updated.
  - New node `macbeth-floor-d1-d2-f19cfcb`: *checked-sober*, scoped to strict $\bar{\alpha}$ and to 1.1, 1.2, Prop. 7 and 1.4. Its `recheck` is this report, and 1.3 is recorded as refuted.
  - New node `macbeth-torus-capstone-e306ae5`: *peer-claimed*, with a `recheck` for Thm 8's value, sign and (5) only.